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# Regional deformational features of the South China Block inferred from Middle Triassic palaeomagnetic data

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## SUMMARY

Middle Triassic red-purple sandstones were sampled at four localities in the northeastern part of the South China Block (SCB) to detect regional deformational features. Most of the samples gave a characteristic remanent magnetization with unblocking temperatures up to 680 °C. The primary nature of this magnetization was ascertained by a positive fold test and a reversal test. The tilt-corrected mean direction from the Yueshan area (30.6°N, 116.8°E), which is located at a distance of 25 km from the eastern part of the Tan-Lu Fault, shows a counter-clockwise deflection of 67.3° with respect to north. However, mean directions from the Nanzhang (31.5°N, 111.7°E), Tongling (30.9°N, 117.9°E) and Nanjing (32.0°N, 118.9°E) areas, which are located between 75 and 90 km from the northeastern boundary of the SCB, indicate a northeasterly deflection in declination, such as $D = 38.2°$, $I = 25.1°$ ($\alpha_{95} = 6.4°$), $D = 32.1°$, $I = 25.2°$ ($\alpha_{95} = 9.7°$) and $D = 34.9°$, $I = 27.8°$ ($\alpha_{95} = 5.4°$), corresponding to palaeopoles of $\lambda = 50.5°$ N, $\phi = 220.5°$ E, $dp/dm = 3.7°/6.9°$, $\lambda = 55.6°$ N, $\phi = 231.8°$ E, $dp/dm = 5.6°/10.4°$ and $\lambda = 53.9°$ N, $\phi = 229.1°$E, $dp/dm = 3.2°/5.9°$, respectively. Comparison with the previously reported Permo-Triassic palaeomagnetic results from the SCB suggests that very consistent palaeopoles are observed within a stable belt up to 1800 km along the northern margin of the SCB (covering the Changxing, Nanjing, Tongling, Nanzhang and Emeishan areas). This type of behaviour implies that an area located between 70 and 225 km from northern margin of the SCB forms a stable platform where very little relative rotation has been detected. However, the inner region of this block, especially the Mesozoic fold-thrust belt, experienced a clockwise rotation of $11.5° \pm 5.0°$ to $30.3° \pm 9.2°$ with respect to the stable outer region later than the Middle Triassic. Areas close to the Tan-Lu Fault were subjected to post-Triassic counter-clockwise rotation by 90° with respect to the stable tectonic belt. We thus conclude that the northern stable belt of the SCB suffered no oroclinal bending during its suturing with the North China Block.

**Key words:** continent, deformation, palaeomagnetism, SCB, tectonics, Triassic.

## 1 INTRODUCTION

One of the most well-debated Palaeozoic–Mesozoic tectonic events within the Asian continent is the collision of the North China Block (NCB) with the South China Block (SCB). The initial contact between these two blocks began during early-Late Permian to early-Middle Mesozoic while a subsequent suturing lasted until the end of the Jurassic (Zhao & Coe 1987; Enkin *et al.* 1992a; Yang *et al.* 1992; Ames *et al.* 1993; Gilder *et al.* 1999; Webb *et al.* 1999; Yokoyama *et al.* 2001). As shown in Fig. 1, these two blocks are separated by the east–west trending Qinling-Dabieshan orogenic belt to the south-southwest and the north-northeast striking Tan-Lu Fault to the east (Chen 1989). Since the deformational features of the Asian continent as a result of its Cenozoic collision with the Indian subcontinent have been well documented (e.g. Tapponnier *et al.* 1982; Houseman & England 1986; Replumaz & Tapponnier 2003), various modes of crustal deformation with a local to continental scale are also anticipated in the SCB and NCB.

Since palaeomagnetism is fundamental to resolving the issue of lithospheric deformation on a local and continental scale, palaeomagnetic data from marginal areas of the SCB are compared with those from its interior to identify tectonic-related deformation within the block. In order to focus on these deformational features, which have been imprinted as a result of the SCB–NCB collision, we present Middle Triassic palaeomagnetic results from the northern margin of the SCB.

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## 2 GEOLOGICAL SETTING AND SAMPLING

As shown in Fig. 1, the SCB is bounded to the west by the Songpan-Ganze orogenic belt and to the north by the Qinling-Dabie orogenic belt. The eastern part of this belt appears to be truncated by the north-northeast striking Tan-Lu fault zone, giving the northern margin of the SCB an L-shaped geometry. The Qinling-Dabie orogenic belt is further divided into north and south orogenic zones (Meng & Zhang 1999). The northern part has been placed into the Palaeozoic era, whereas the southern part is regarded as Mesozoic with abundant Triassic granite and Late Palaeozoic to Early Mesozoic metamorphic signatures (Mattauer *et al.* 1985; Zhai *et al.* 1998; Sun *et al.* 2002). The Mesozoic orogenic belt, which is represented by the Xiangfan-Guangji Fault, marks the northern limit of the SCB (Wang *et al.* 1992; Cong 1996; Wang *et al.* 1998; Liu *et al.* 2003).

The oldest rocks in the SCB are Archean. Weakly metamorphosed Upper Silurian to Lower Triassic platform carbonates form the basement rocks in the northern part of the SCB (BGMRJS 1977; BGMRHN 1988; BGMRHB 1990; BGMRSC 1991). The Palaeozoic strata are overlain by Early and Middle Triassic shallow marine sediments. Continental sedimentation in the study area started after the Late Triassic. Conformably placed Jurassic and Lower Cretaceous strata were folded together with older strata in the northwest of the Mesozoic fold-thrust belt (MFTB), suggesting Cretaceous deformational activities in the area. The axes of the folds and thrusts strike towards the northeast in the central part of the SCB, but their trend changes to east-northeast in the north. In the foreland of the Qinling-Dabie orogenic belt, the axes of folds and thrusts follow an east-southeast trend.

The areas involved in the present study are located to the south of the L-shaped Qinling-Dabie–Tan-Lu zone, the northeastern part of the SCB. We focus our palaeomagnetic study on the Middle Triassic ($T_2$) purplish-red sandstone formations (including the Badong, Tongtoujian and Huangmaqing formations), where samples were collected fromt four different areas: Nanzhang (31.5°N, 111.7°E; Hubei Province), Tongling (30.9°N, 117.9°E; Anhui Province), Nanjing (32.0°N, 118.9°E; Jiangsu Province) and Yueshan (30.6°N, 116.8°E; Anhui Province). These sampling areas are located at 90, 75, 70 and 25 km, respectively, from the northern margin of the SCB. The Badong Formation of Hubei Province corresponds to the Tongtoujian Formation of Anhui Province and the Huangmaqing Formation of Jiangsu Province.

### 2.1 The Nanzhang area

Triassic strata in the Nanzhang area are divided into four units in ascending order: the Lower Triassic Daye and Jialingjiang formations, the Middle Triassic Badong Formation and the Upper Triassic Jiuligang Formation (Fig. 2a). The Middle Triassic Badong Formation is about 1087 m thick and consists of four different stratigraphic members (BGMRHB 1990). The lower two members have been assigned to an Anisian age and the upper two members to a Ladinian age. As shown in Fig. 2(a), red-purple sandstones of all four members were sampled from 22 sites (206 samples) 40 km southwest of Nanzhang County (31.5°N, 111.7°E). The strata dip nearly north- or southward and the angle of dip ranges between 16° and 44°. Nine out of 22 sites (HB49–HB57) were sampled from both limbs of an anticline.

### 2.2 The Tongling area

The Middle Triassic rocks of the Tongling area have been named the Tongtoujian Formation by BGMRAH (1987). The Middle Triassic

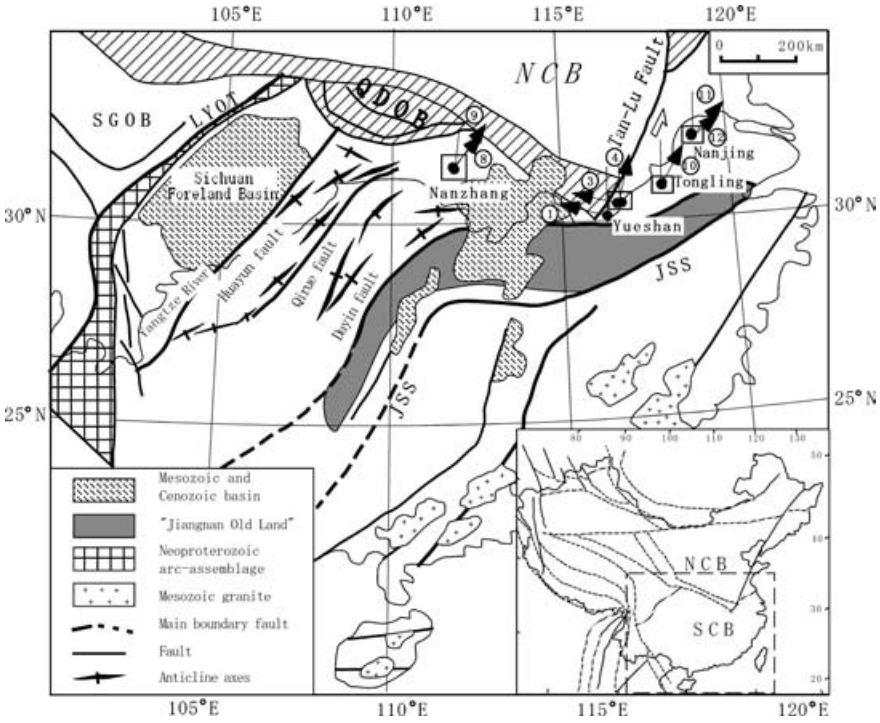



**Figure 1.** Simplified tectono-geological map of eastern China after Yan *et al.* (2003): NCB, North China Block; SCB, South China Block; QDOB, Qinling-Dabie orogenic belt; SGOB, Songpan-Guanze fold belt; LYOT, Longmenshan-Yanyuan overthrust fault; JSS; Jingshan-Shaoxing suture. The available Middle Triassic declinations from the northern margin of the SCB are shown by arrows at the studied locations. The numbers adjacent to arrows represent the corresponding localities in Table 2.

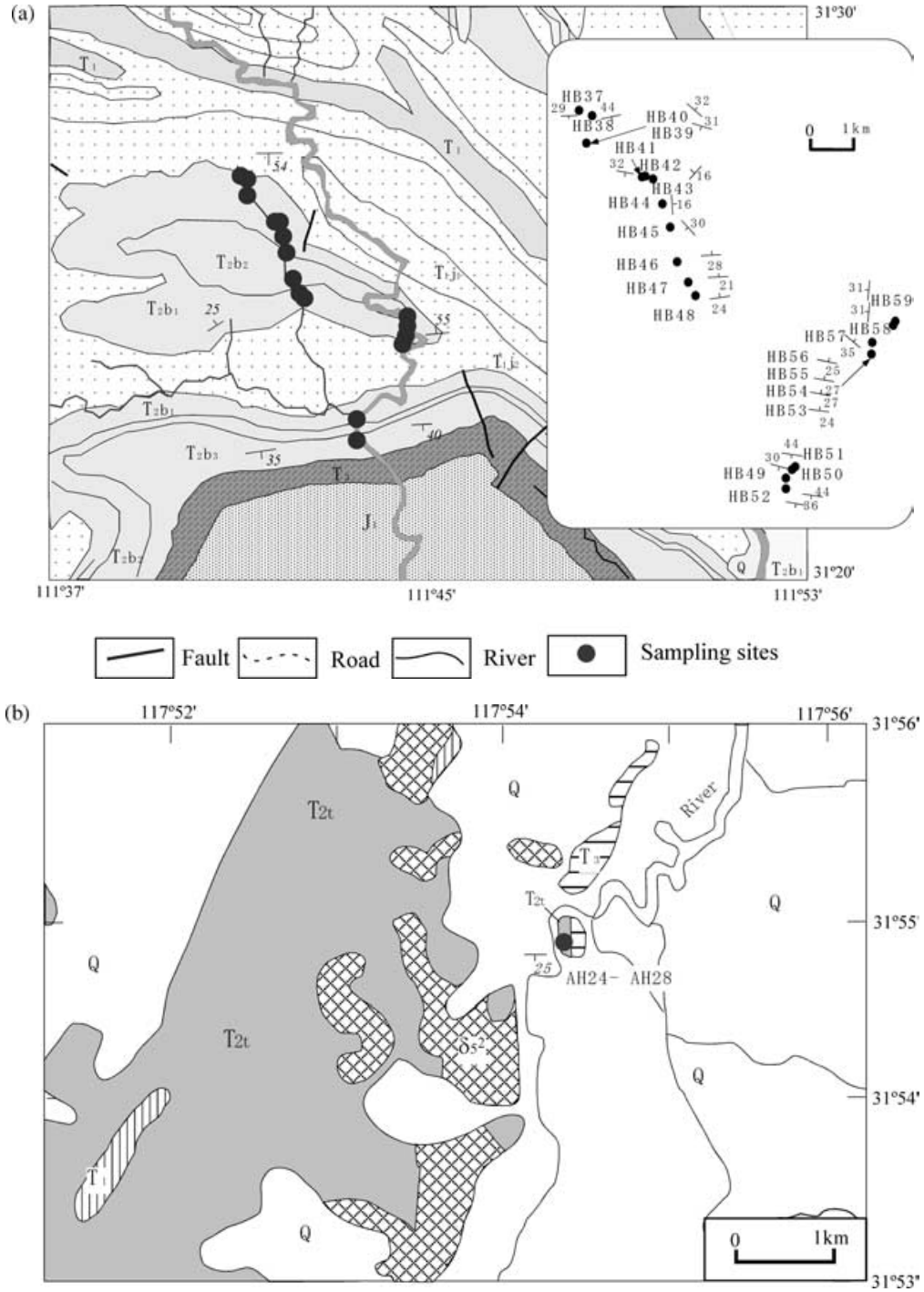



**Figure 2.** Geological maps of the study area (after BGMRJS 1977; BGMRAH 1987; BGMRHB 1986): (a) Nanzhang, (b) Tongling, (c) Nanjing, (d) Yueshan. Sampling sites are shown by closed circles. For the Nanzhang area, strike/dip orientations of strata at sampling sites are shown in the inset, while for the Tongling and Yueshan areas orientation coordinates are mentioned adjacent to each sampling site. Key: $\delta 5^2$, Yanshanian diorite; $\delta\mu$, diorite porphyrite; $\eta o\pi$, quartz monzanitic porphyry; $\chi$, alkaline rock; $\gamma$, Granite; $\delta\mu_5$, Yanshanian diorite porphyrite; S, Silurian; BD, before Devonian; D, Devonian; C, Carboniferous; P, Permian; T, Triassic; J, Jurassic; K Cretaceous; E, Palaeocene to Eocene; Q, Quaternary; W, water; 1, Early; 2, Middle; 3, Upper; $T_1j_{1,2}$, the Jialingjiang Formation; $T_2b_{1,2,3}$, the Badong Formation; $T_2t$, the Tongtoujian Formation; $T_2b$, the Biandanshan Formation; $T_2h$, the Huangmaqing Formation. $T_{2x}$, the Xujiashan Formation; $T_{2c}$, the Changbomen Formation; $T_{3f}$, the Fanjitang Formation.

age is well constrained by the presence of plant fossils and a bivalve assemblage. In the Tongling area (Fig. 2b), this formation conformably overlies dolomite and karst breccia series of the Dongmaanshan Formation, and is overlain by the Late Triassic Lalitou Formation (which is made up of grey to grey-black sandstone, siltstone, carbonaceous shale and unstable coal beds). In Chaoshan village of Tongling County (30.9°N, 117.9°E), we collected 45 samples (five sites) from Middle Triassic red-purple sandstone of the Tongtoujian Formation. The sampling strata generally dip up to 30° to the east.

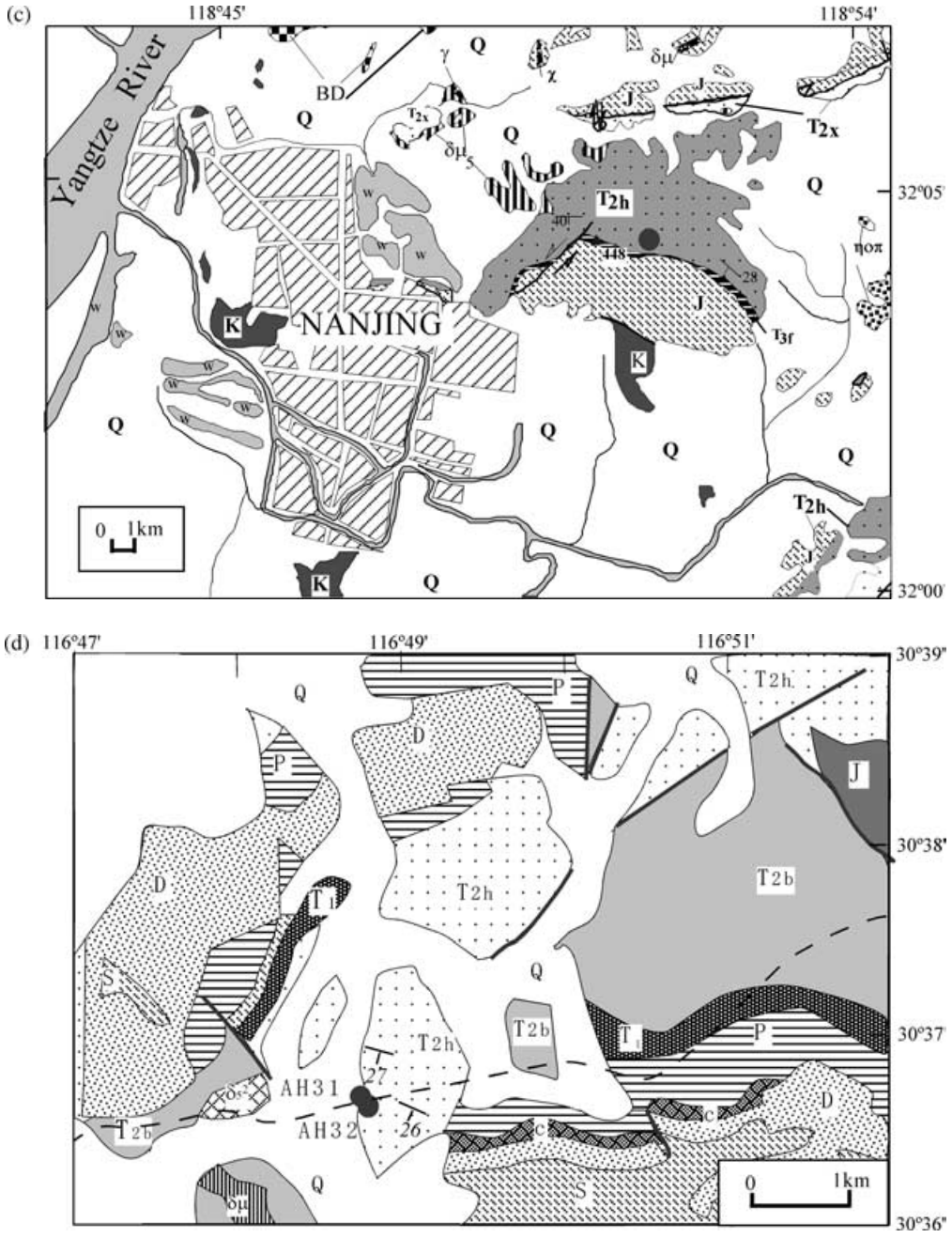



**Figure 2.** (*Continued*.)

### 2.3 The Nanjing area

The type section of the Middle Triassic (Anisian stage) Huangmaqing Formation extends from Xiawuqi village to Zijinshan Hill on the northeast side of the Nanjing City (32.0°N, 118.9°E). The Huangmaqing Formation conformably overlies the Zhouchongcun Formation, and is in turn conformably overlain by the coal-bearing Upper Triassic Fanjiatang Formation (BGMRJS 1977; BGMRAH 1987). The Huangmaqing Formation consists of clastic rocks such as purplish-red siltstone, sandy shale and fine-grained sandstones. A total of 181 cores of fine-grained purplish-red sandstones were collected over a stratigraphic thickness of 224.5 m. The strata in the sampled area generally dip between 16° and 26° toward the south.

### 2.4 The Yueshan area

In Yueshan County, the sedimentary strata are generally characterized by Middle Triassic red-purple sandstone of the Huangmaqing Formation (BGMRAH 1987). Due to limited exposure and drilling difficulties in the area, only 20 samples at two sites could be collected at Dengxi village (30.6°N, 117.0°E). The bedding attitude at this locality generally dips towards the south.

## 3 FIELD AND LABORATORY PROCEDURES

Palaeomagnetic samples were cored using a portable drill machine and oriented with a magnetic compass. In Nanzhang, Tongling and

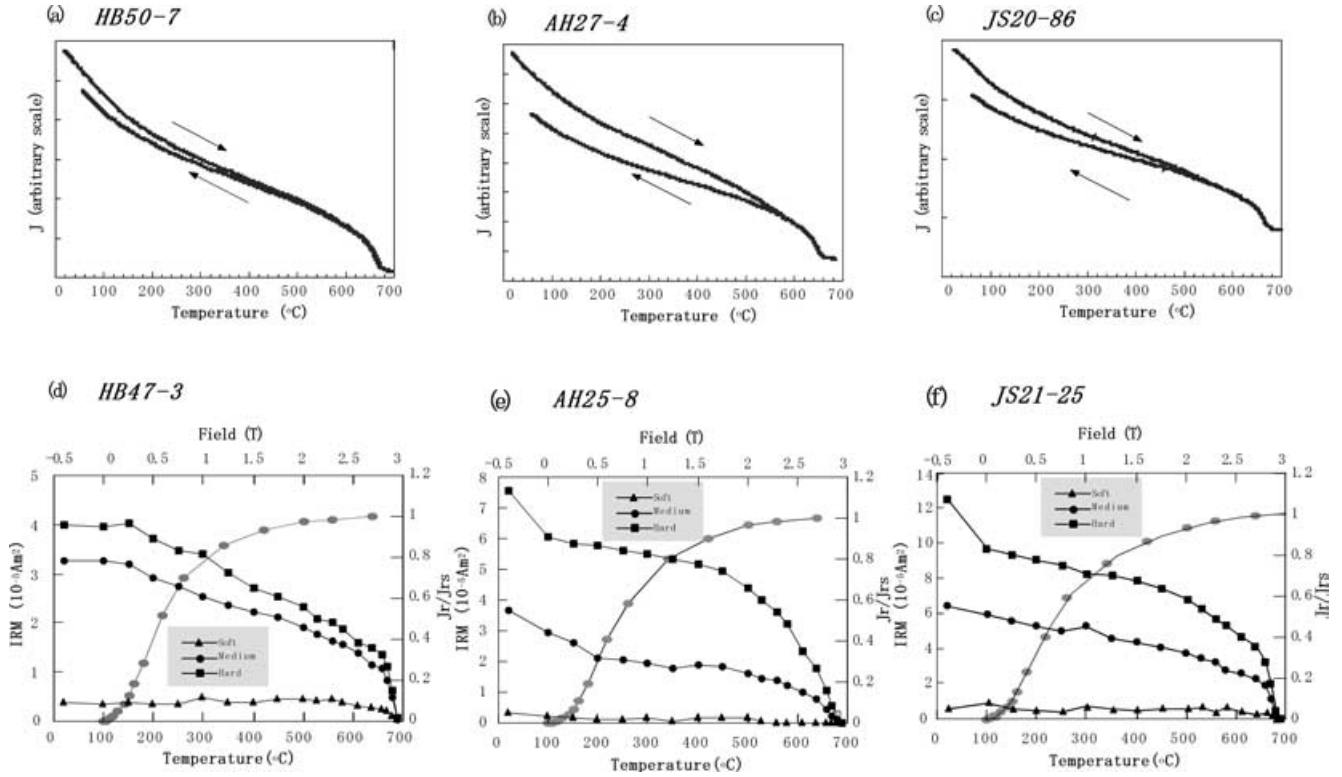



**Figure 3.** (a), (b), (c) Thermomagnetic curves for representative specimens indicating a Curie point of haematite at about 685 °C. High-field magnetization is conventionally in arbitrary units. (d), (e), (f) IRM acquisition curves (open circles) for the representative specimens, and thermal demagnetization of composite IRMs (see details in text). HB50-7 and HB 47-3 are samples from the Nanzhang area, HB27-4 and AH25-8 are from the Tongling area and JS21-31 and JS21-25 are from the Nanjing area.

Yueshan areas, seven to 12 samples were collected at each site over a distance of 15 m. In the Nanjing area, one core sample was collected from each layer where the exposure consists of many layers with a thickness of tens to a hundred centimetres. Location coordinates of the sampling areas were determined by a portable navigational instrument (Sony Pyxis). The present declination value at each sampling locality was evaluated from the International Geomagnetic Reference Field, 1995 (IAGA Division V, Working Group 8 1995).

In the laboratory, 2.2 cm long specimens were sliced from each sample. The natural remanent magnetization (NRM) of each specimen was measured using a cryogenic magnetometer (2G Enterprises) at the Palaeomagnetic Laboratory, Kobe University. All specimens were subjected to stepwise thermal demagnetization using a Natsuhara TDS-1 thermal demagnetizer. The residual field of this demagnetizer was less than 5 nT. On average, 17 heating steps were used up to 690 °C, where temperature interval was as little as 5 °C from 670 to 690 °C. The bulk magnetic susceptibility of each specimen was measured after each demagnetization step in order to monitor magnetic mineralogical alteration during progressive heating. The magnetic behaviour of each specimen after the demagnetization process was plotted on a Zijderveld diagram (Zijderveld 1967). Principal component analysis (Kirschvink 1980) was used to identify different magnetic components. Site-mean directions were calculated by Fisherian statistics (Fisher 1953).

Progressive acquisition of isothermal remanent magnetization (IRM) was performed up to a maximum field of 2.7 T using a 2G pulse magnetizer. Thermal demagnetization of the composite IRMs (Lowrie 1990) was carried out using a Natsuhara TDS-1 demagnetizer. For the identification of ferromagnetic minerals, several specimens were subjected to thermomagnetic analysis using an automatic recording thermomagnetic balance at Kyoto University.

## 4 ROCK MAGNETIC PROPERTIES

The typical thermomagnetic curves (Figs 3a–c) suggest a predominance of haematite as the magnetic carrier in the studied specimens from all four areas. The heating phase of the curve shows a very clear fall at about 685 °C. The IRM acquisition curves show that a saturation limit is not achieved up to a maximum field of 2.7 T (Figs 3e–f), indicating the dominance of high-coercivity minerals such as haematite and/or goethite. The presence of haematite is very clearly shown by thermal demagnetization of the composite IRMs. The presence of goethite, however, is not indicated.

## 5 PALAEOMAGNETIC RESULTS

### 5.1 The Nanzhang area

Initial NRM intensities of the studied specimens varied from $5.1 \times 10^{-5}$ to $1.0 \times 10^{-2}$ A m$^{-1}$. In most of the 22 sites, a partial thermal demagnetization procedure revealed two remanent components (Figs 4a–f). The low-temperature component is unblocked between

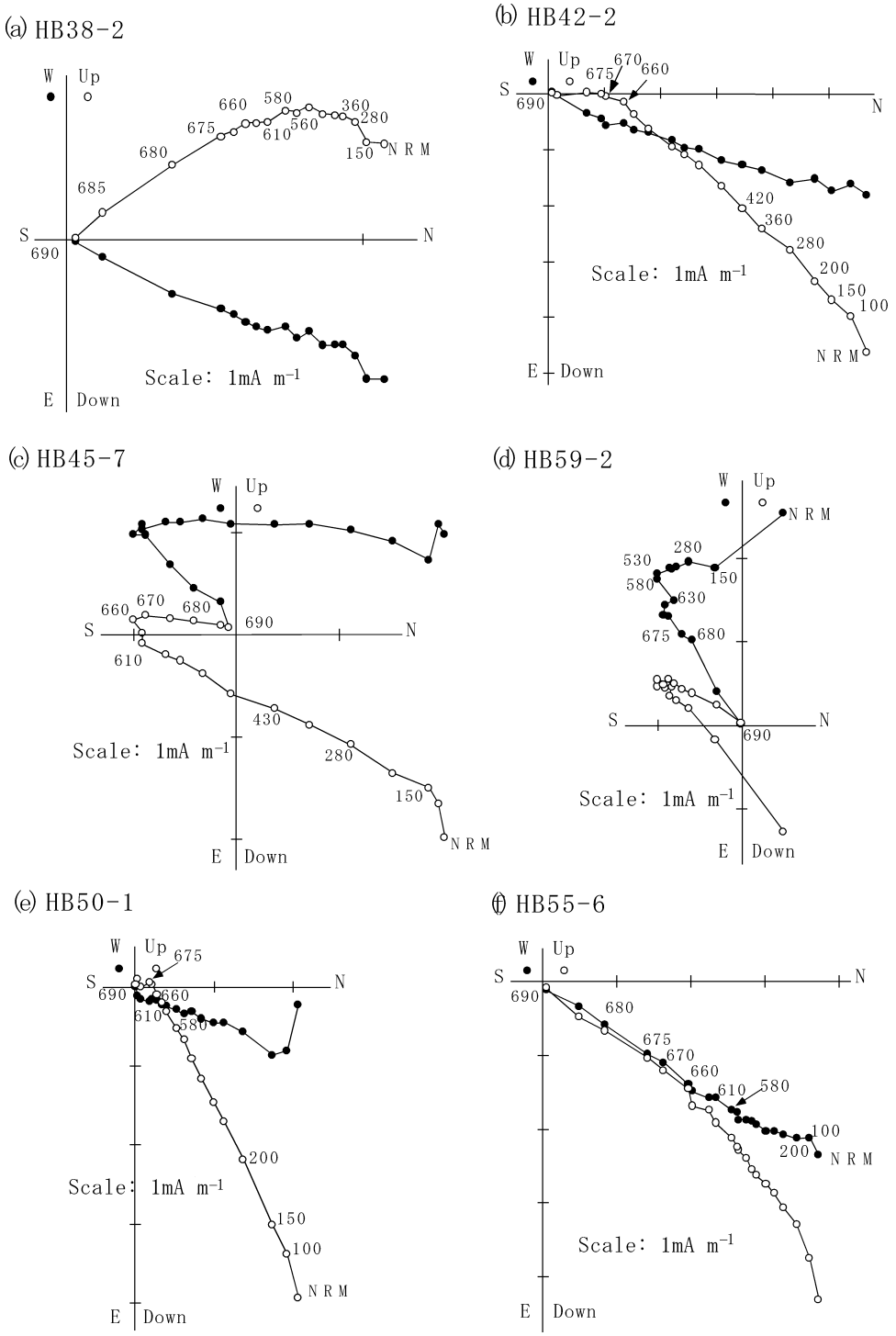



**Figure 4.** Vector endpoint diagrams for representative specimens after thermal demagnetization. The solid/open symbols indicate horizontal/vertical projections. All results are in geographical coordinates. The numbers adjacent to the decay plot are the heating steps in °C. (a)–(f), (g)–(j), (k)–(p) and (q, r) Results of specimens for the Nanzhang, Tongling, Nanjing and Yueshan areas, respectively.

360 and 660 °C, which generally gave a north-northeast direction with moderate to steep-downward inclination (in geographical coordinates). In geographical coordinates, the formation-mean direction for this component is $D = 13.7°$, $I = 50.3°$ ($k = 25.2$, $\alpha_{95} = 6.3°$, $N = 22$). The fold test (McFadden & McElhinny 1990) is negative at the 99 per cent confidence limit, indicating that the lower-temperature component in the red beds of the Nanzhang area is clearly post-folding in origin.

The high-temperature component is separated in a range between 610 and 690 °C, indicating haematite as a magnetic carrier. This characteristic component has either a normal or reversed polarity direction. Seventeen out of 22 sites predominantly gave normal

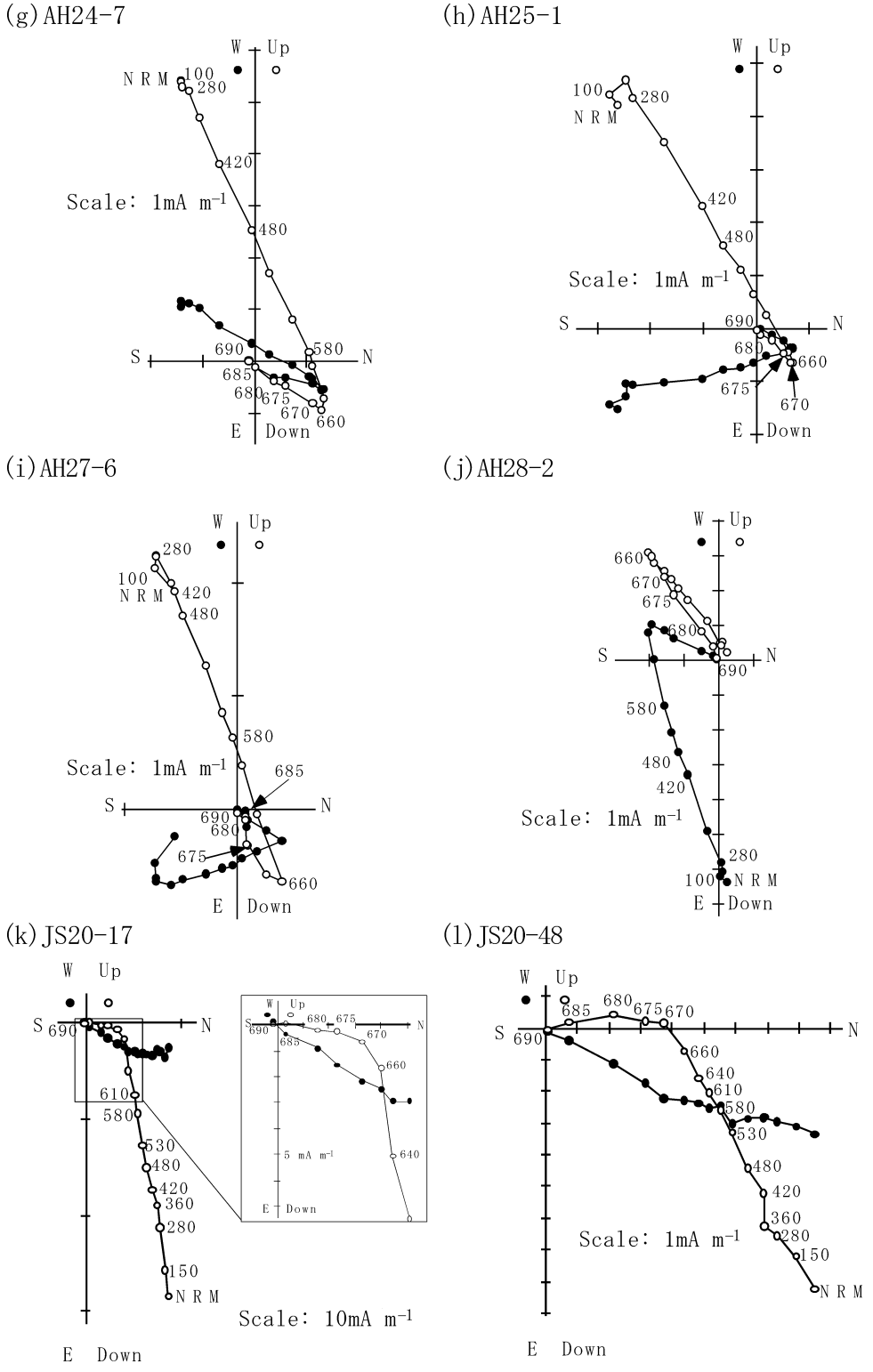



**Figure 4.** (*Continued*.)

polarity directions, whereas the remaining five sites revealed reversed polarity directions (Fig. 5a). The site mean direction of one site (HB39) is discarded due to an unreliable dip direction. Thus, the formation-mean direction of 21 sites is $D = 43.0^\circ$, $I = 22.4^\circ$ ($k = 8.8$, $\alpha_{95} = 11.4^\circ$) in geographical coordinates and $D = 38.2^\circ$, $I = 25.1^\circ$ ($k = 25.4$, $\alpha_{95} = 6.4^\circ$) in stratigraphic coordinates. The optimal concentration of the direction-correction (DC) tilt test (Enkin 2003) is achieved at 90 $\pm$19 per cent, which indicates a positive tilt test. According to the method of McFadden & McElhinny (1990), the fold test is also positive at the 99 per cent confidence level, where the calculated value ($\xi_1$) is 8.954 in geographical coordinates and 0.818 in stratigraphic coordinates, while the critical value ($\xi$) is 7.483 at the 99 per cent confidence level.

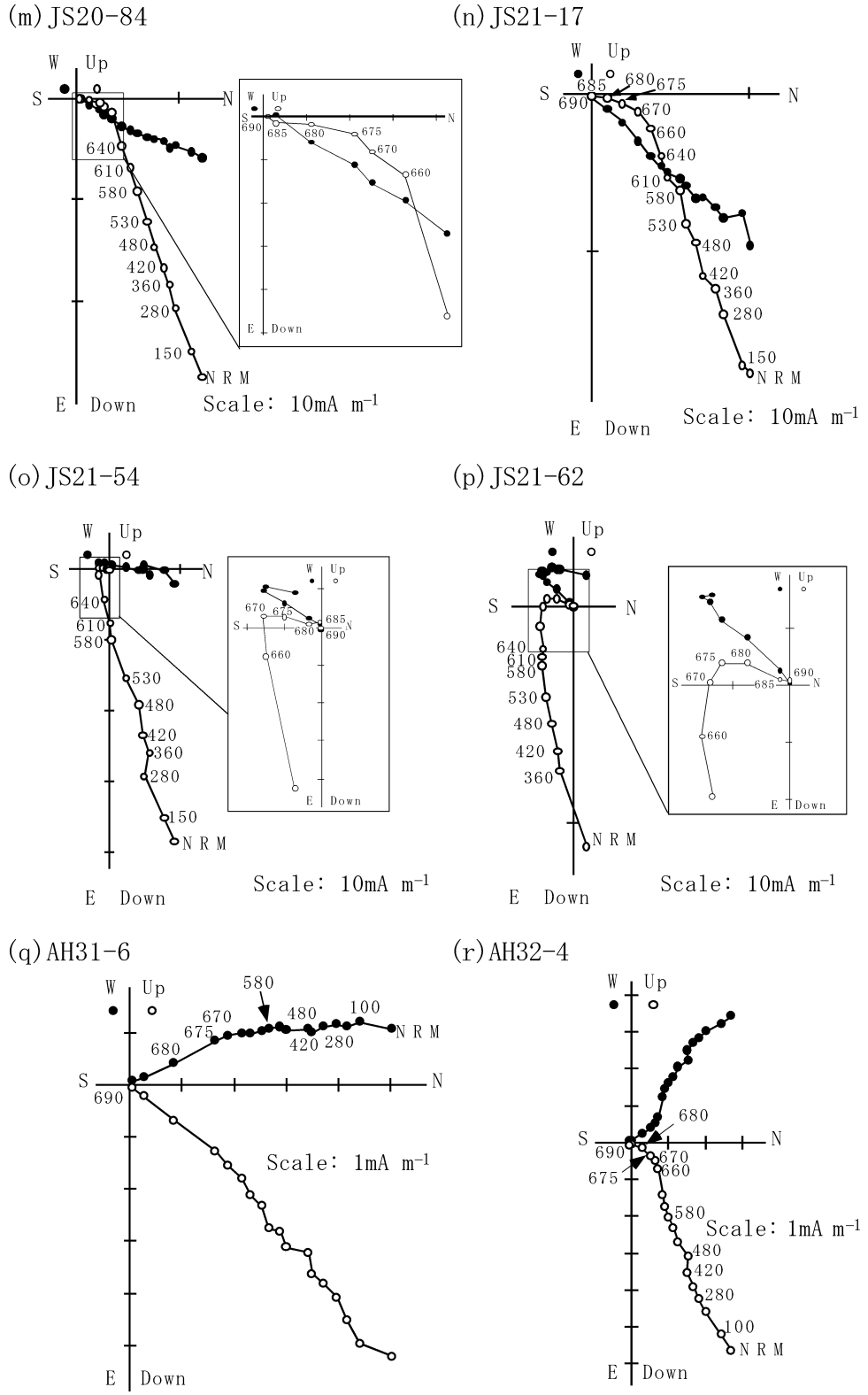



**Figure 4.** (*Continued*.)

The tilt-corrected mean direction of normal polarity sites is $D = 39.0^\circ$, $I = 25.1^\circ$ ($k = 30.9$, $\alpha_{95} = 6.7^\circ$, $N = 16$), while that of reversed polarity sites is $D = 215.5^\circ$, $I = -25.0^\circ$, ($k = 13.5$, $\alpha_{95} = 21.6^\circ$, $N = 5$) (Table 1). Tilt-corrected mean directions of both polarities passed a reversal test at the 95 per cent confidence level (McFadden & McElhinny 1990). The angular distance between the tilt-corrected mean directions of normal and reversed polarities is 3.2°, which is less than the critical angle of 15.6°. The positive fold and reversal tests thus suggest that the high-temperature component obtained from the Middle Triassic Badong Formation is the characteristic palaeomagnetic direction for the Nanzhang area. This is in good agreement with the northeasterly mean direction

**Table 1.** Middle Triassic palaeomagnetic results from the Nanzhang, Tongling, Nanjing and Yueshan areas of the South China Block. Magnetization direction of high temperature component are listed. N(n) is the number of samples measured (number of samples used for calculation); (d,c), directions obtained by principal component analysis/great circles, Dec. and Inc. are declination and inclination, respectively; k is the estimate of the precision parameter (Fisher, 1953); $\alpha_{95}$ is the radii of the cone of 95 per cent confidence. Strikes of bedding are values before the subtraction of local geomagnetic declination. Local declination values are calculated for the present local geomagnetic declination using the Definitive Geomagnetic Reference Field of 1995 (IAGA Division V, Working Group 8, 1995). N, R and I represent normal, reversed and intermediate polarity directions, respectively. A site (HB39) denoted by an asterisk is excluded from formation mean calculation because of problem with dip direction.

| Site no | $n/N$ or $n(d, c)$ | *In situ* | | Tilt corrected | | $\kappa$ | $\alpha_{95}$ | Polarity |
|---|---|---|---|---|---|---|---|---|
| | | Dec. (°) | Inc.(°) | Dec.(°) | Inc.(°) | | | |
| Nanzhang (31.5°N, 111.7°E) | | | | | | | | |
| HB37 | 8/10 | 47.4 | 2.5 | 52.1 | 19.5 | 17.7 | 13.5 | N |
| HB38 | 7/7 | 237.5 | 29.6 | 222.4 | 10.0 | 3.2 | 39.9 | R |
| HB39* | 10/12 | 74.7 | 73.3 | 160.1 | 63.3 | 21.9 | 10.6 | N |
| HB40 | 2/6 | 60.4 | 69.2 | 47.4 | 38.4 | — | — | N |
| HB42 | 6/6 | 26.5 | 7.1 | 24.8 | 22.9 | 97.6 | 6.8 | N |
| HB43 | 6/9 | 37.6 | 15.3 | 33.4 | 14.0 | 52.9 | 9.3 | N |
| HB44 | 7/10 | 37.2 | 13.5 | 33.1 | 24.3 | 24.9 | 12.3 | N |
| HB45 | 7/10 | 218.3 | −4.8 | 216.9 | −34.6 | 16.0 | 15.6 | R |
| HB46 | 10/13 | 220.5 | −43.7 | 208.5 | −22.1 | 29.2 | 9.1 | R |
| HB47 | 6/8 | 53.5 | 38.7 | 42.6 | 26.2 | 32.9 | 11.9 | N |
| HB48 | 8/10 | 45.2 | 44.1 | 31.7 | 28.3 | 65.7 | 6.9 | N |
| HB49 | 7/10 | 31.0 | −0.6 | 38.8 | 38.1 | 19.8 | 13.9 | N |
| HB50 | 7/8 | 27.7 | −2.9 | 29.5 | 25.9 | 15.9 | 15.6 | N |
| HB51 | 9/10 | 36.6 | 9.8 | 51.7 | 44.9 | 175.6 | 3.9 | N |
| HB52 | 2/7 | 57.7 | −18.4 | 54.6 | 5.7 | — | — | N |
| HB53 | 8/10 | 68.2 | 45.7 | 51.9 | 32.0 | 103.4 | 4.8 | N |
| HB54 | 8/10 | 36.9 | 48.0 | 28.3 | 23.5 | 26.4 | 10.2 | N |
| HB55 | 7/10 | 38.5 | 43.5 | 30.7 | 19.4 | 32.8 | 9.8 | N |
| HB56 | 5/9 | 24.9 | 39.6 | 21.7 | 15.4 | 136.8 | 5.7 | N |
| HB57 | 8/8 | 65.5 | 48.8 | 56.1 | 16.5 | 34.7 | 9.5 | N |
| HB58 | 7/8 | 233.1 | −27.8 | 214.1 | −49.0 | 42.3 | 9.4 | R |
| HB59 | 8/10 | 223.3 | −8.4 | 215.1 | −27.5 | 19.0 | 12.1 | R |
| Mean | 21 | 43.0 | 22.4 | 38.2 | 25.1 | 8.8 | 11.4 | |
| | | | | | | 25.4 | 6.4 | |
| Tongling (30.9°N, 117.9°E) | | | | | | | | |
| AH24 | 7/8 | 18.6 | 35.7 | 31.8 | 22.6 | 40.7 | 9.6 | N |
| AH25 | 8/9 | 21.9 | 43.4 | 36.9 | 28.6 | 62.9 | 7.0 | N |
| AH26 | 11/12 | 5.1 | 29.2 | 16.7 | 21.7 | 22.5 | 9.8 | N |
| AH27 | 8/8 | 17.4 | 50.9 | 38 | 32.2 | 34.2 | 9.6 | N |
| AH28 | 7/8 | 208.6 | −41.1 | 220.8 | −19.9 | 48.6 | 8.7 | R |
| Mean | 5 | 17.9 | 40.3 | 32.1 | 25.2 | 58.1 | 10.1 | |
| | | | | | | 63.1 | 9.7 | |
| Nanjing (32.0°N, 118.9°E) | | | | | | | | |
| JS1 | 20(8, 12) | 31.2 | 15.5 | 36.2 | 32.8 | 19.3 | 7.7 | N |
| JS2 | 20(10, 10 | 25.1 | 6.9 | 26.7 | 21.5 | 25.2 | 6.7 | N |
| JS3 | 20(4, 16) | 37.5 | 12.0 | 41.0 | 30.8 | 18.2 | 8.0 | N |
| JS4 | 20(8, 12) | 35.2 | 12.4 | 39.1 | 33.1 | 32.4 | 5.9 | N |
| JS5 | 20(3, 17) | 38.8 | 19.6 | 42.1 | 33.5 | 43.5 | 5.1 | N |
| JS6 | 20(3, 17) | 30.8 | −2.5 | 31.4 | 17.3 | 13.7 | 9.3 | N |
| JS7 | 20(7, 13) | 210.6 | −6.9 | 212.8 | −25.1 | 36.6 | 5.6 | R |
| JS8 | 23(14, 9) | 208.5 | −5.9 | 210.6 | −27.8 | 32.1 | 5.5 | R |
| Mean | 8 | 32.2 | 9.6 | 34.9 | 27.8 | 98.3 | 5.6 | |
| | 8 | | | | | 107.6 | 5.4 | |
| Yueshan (30.6°N, 116.8°) | | | | | | | | |
| AH31 | 8/10 | 329.3 | 37.1 | 306.9 | 53.9 | 39.7 | 8.9 | N |
| | | | | | | 39.7 | 8.9 | |
| AH32 | 7/10 | 300.0 | 44.5 | 279.0 | 42.8 | 10.5 | 19.6 | N |
| | | | | | | 12.5 | 17.8 | |
| Mean | 15 | 317.0 | 41.3 | 292.7 | 49.7 | 13.5 | 10.8 | |
| | | | | | | 15.5 | 10.0 | |

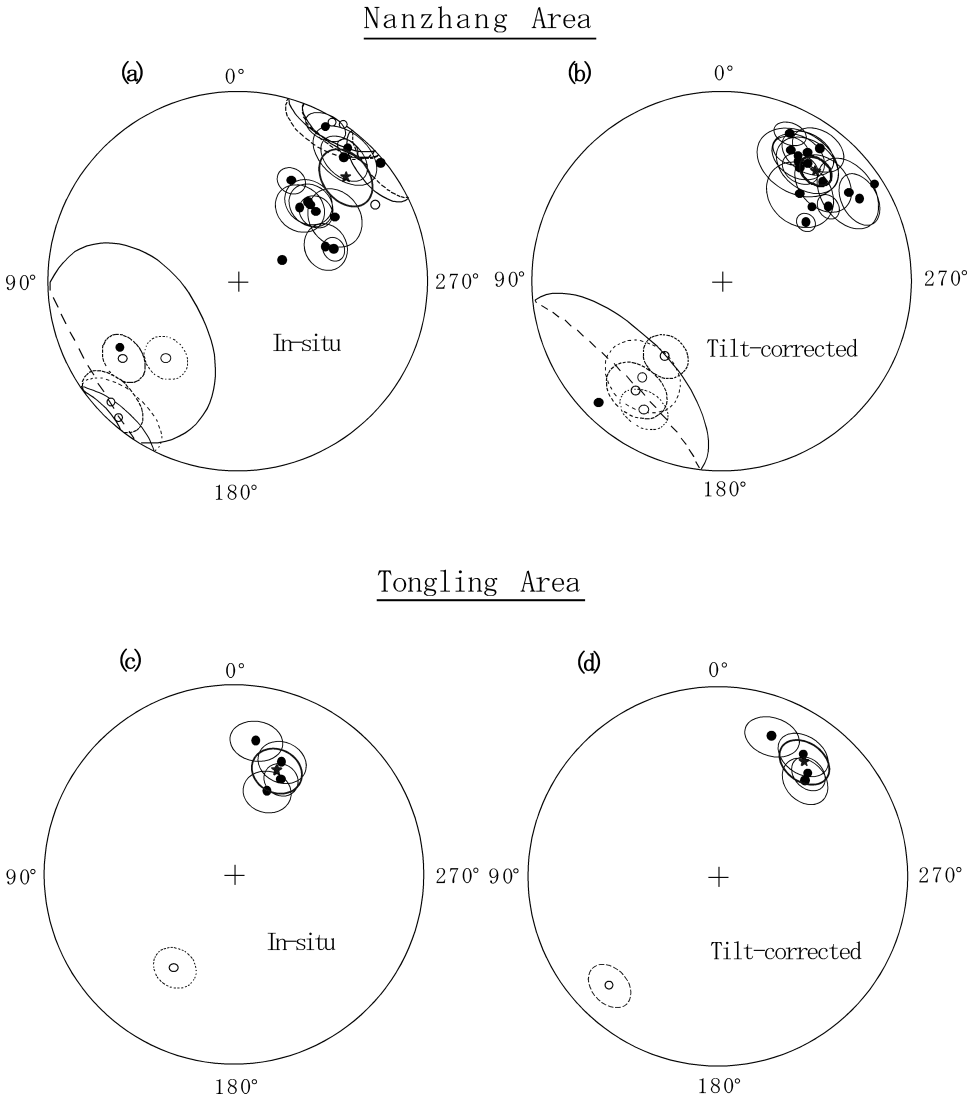



**Figure 5.** Equal-area projections of high-temperature components. (a), (b) Results for the Nanzhang area. (c), (d) Results for the Tongling area. (e), (f) Results for the Nanjing area. (g), (h) Results for the Yueshan area. Site mean directions with 95 per cent circles of confidence are drawn before and after tilt correction. Solid (open) circles represent projection onto the lower (upper) hemisphere. Stars represent mean directions.

($D = 38.2^\circ$) reported by Huang & Opdyke (1997) from the Liujiatai and Maoping sections of the Badong Formation, the Jingmeng-Dangyang Basin.

### 5.2 The Tongling area

Initial NRM intensities of 45 specimens vary from $1.2 \times 10^{-4}$ to $1.4 \times 10^{-3}$ A m$^{-1}$ and the majority are in a range from (5–8) $\times$ $10^{-4}$ A m$^{-1}$. After the removal of the lower-temperature component between 300 and 660 °C, the high-temperature component appeared between 660 and 685 °C. This high-temperature component from four out of five sites (AH24, AH25, AH26 and AH27), in which a linear decay toward the origin is observed (Figs 4g–j), shows a northeastward direction with a downward inclination (Fig. 5b). *The* in situ mean direction for these four sites is $D = 15.3^\circ$, $I = 40.0^\circ$ ($k = 53.6$, $\alpha_{95} = 12.7^\circ$). Specimens from the remaining site (AH28) gave a southwestward direction with upward inclination (i.e. $D = 208.6^\circ$, $I = -41.1^\circ$, $k = 48.6$, $\alpha_{95} = 8.7^\circ$, $n = 7$). Although both the fold tests are inconclusive, the optimal concentration estimated according to the DC tilt test (Enkin 2003) is achieved at 129 per cent. After 100 per cent tilt correction, four normal polarity sites gave a mean direction ($D = 30.6^\circ$, $I = 26.5^\circ$, $\alpha_{95} = 11.6^\circ$, $N = 4$ sites). The remaining seven specimens from one site gave a reversed polarity direction ($D = 220.8^\circ$, $I = -19.9^\circ$, $\alpha_{95} = 8.7^\circ$). These antipodal mean directions are not statistically distinguishable from each other.

### 5.3 The Nanjing area

Initial NRM intensities are within a range of $6.8 \times 10^{-3}$ to $5.7 \times 10^{-2}$ A m$^{-1}$. Progressive thermal demagnetization generally revealed three magnetic components (Figs 4k–p). After the removal of an unstable remanent component around 350 °C, an intermediate-temperature component with northward declination and steep inclination (*in situ*) is isolated by 670 °C in 163 samples. A mean direction of $D = 26.7^\circ$, $I = 73.5^\circ$, ($n = 163$, $\alpha_{95} = 1.3^\circ$) is obtained for this component before tilt correction. The fold test is negative at the 95 per cent confidence level, suggesting a post-folding origin for this component. The very steep inclination of this component is

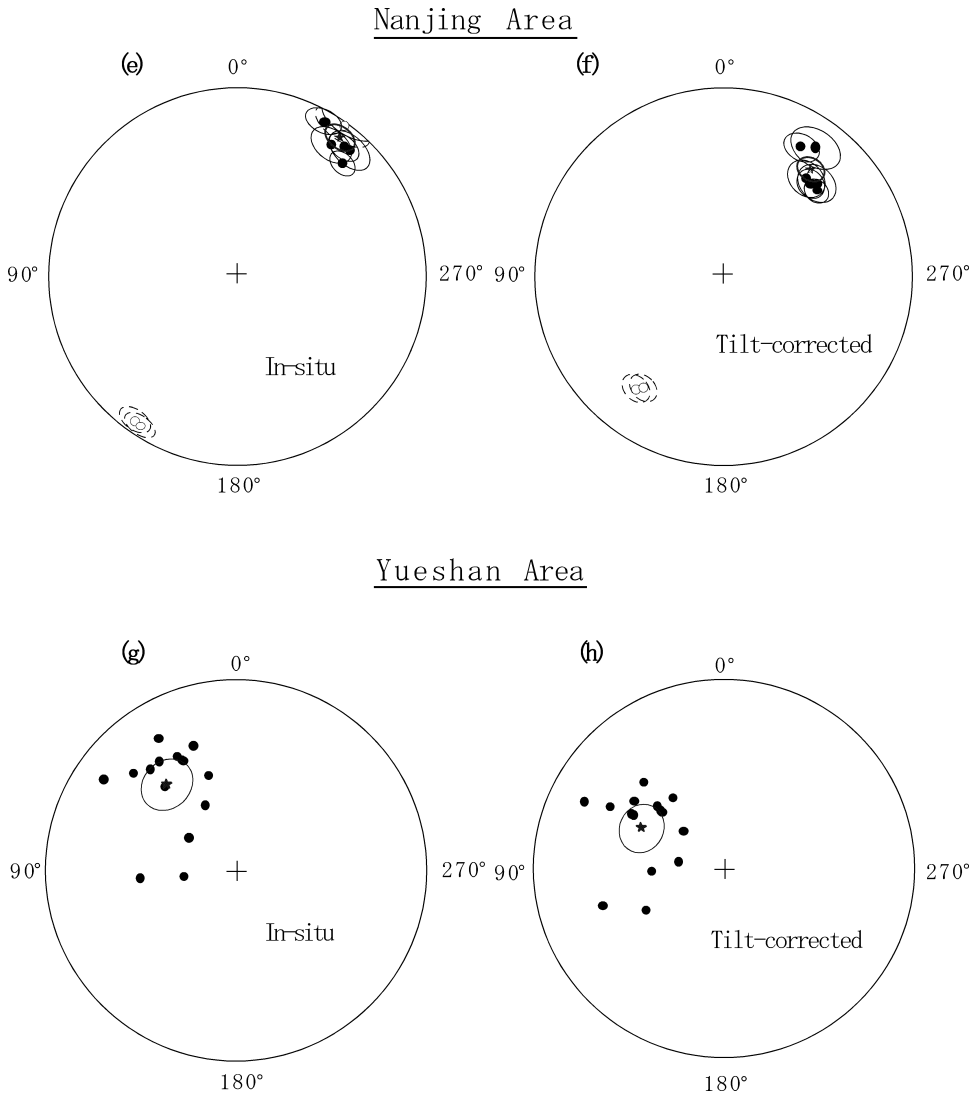



**Figure 5.** (*Continued*.)

nearly identical to one reported by Opdyke *et al.* (1986) from the same formation in the Nanjing area.

The most stable component is unblocked between temperature levels of 675 and 690 °C. Out of 181 specimens, 57 revealed a linear decay of this component to the origin (Fig. 4k, I), whereas 106 specimens show a curvilinear decay to the origin on a demagnetization path between 675 and 690°C (Figs 4m–p). The remaining 18 specimens with erratic directions were discarded for further analysis.

Remanent directions of the 57 specimens were determined by principal component analyses (Kirschvink 1980), which show the presence of both normal and reversed polarities against the stratigraphic thickness of 224.5 m (Fig. 6). Ascending from the lowest part of the sequence, a few layers of mixed polarity are followed by a zone of reversed polarity up to 160 m. The overlying layers are exclusively occupied by a normal polarity zone.

Since the bedding attitude varies only slightly throughout the stratigraphic sequence, all the strata were thus divided into eight sites (JS1 to JS8). The purpose was to estimate site and formation mean directions (Table 1 and Fig. 5). Each site-mean direction was calculated by combining the linear and great circle fitting using the statistical method of McFadden & McElhinny (1988). The optimal concentration of the DC tilt test (Enkin 2003) is achieved at the 93 per cent level while a precision parameter ($k$) increases from 98.3 to 107.6 (Fig. 5) after 100 per cent untilting. The very small change in bedding attitude probably obscured the significance of the fold tests. However, the mean directions of normal and reversed polarities (after 100 per cent unfolding) are not distinguishable from each other. Six out of eight normal polarity sites (JS1 to JS6) give a mean direction of $D = 35.8^\circ$, $I = 28.3^\circ$, $k = 83.2$, $\alpha_{95} = 7.3^\circ$ ($N = 6$) and the remaining two reversed polarity sites (JS7 and JS8) give a mean direction of $D = 211.7^\circ$, $I = -26.7^\circ$, $k = 34.5$, $\alpha_{95} = 3.9^\circ$, $n/N = 40$ specimen/2 sites). The angular distance between these antipodal directions is only 4.1°. The northeasterly formation-mean direction presented here ($D = 34.9^\circ$, $I = 27.8^\circ$, $\alpha_{95} = 5.4^\circ$, $N = 8$) is almost similar to that reported by Opdyke *et al.* (1986) from the same section of the Huangmaqing Formation ($D = 44.8^\circ$), although the latter was obtained from a mediocre incomplete demagnetization of 23 samples.

### 5.4 The Yueshan area

The initial NRM intensity ranged between 3.1 and $8.0 \times 10^{-3}$ A m$^{-1}$. Progressive thermal demagnetization revealed two remanent

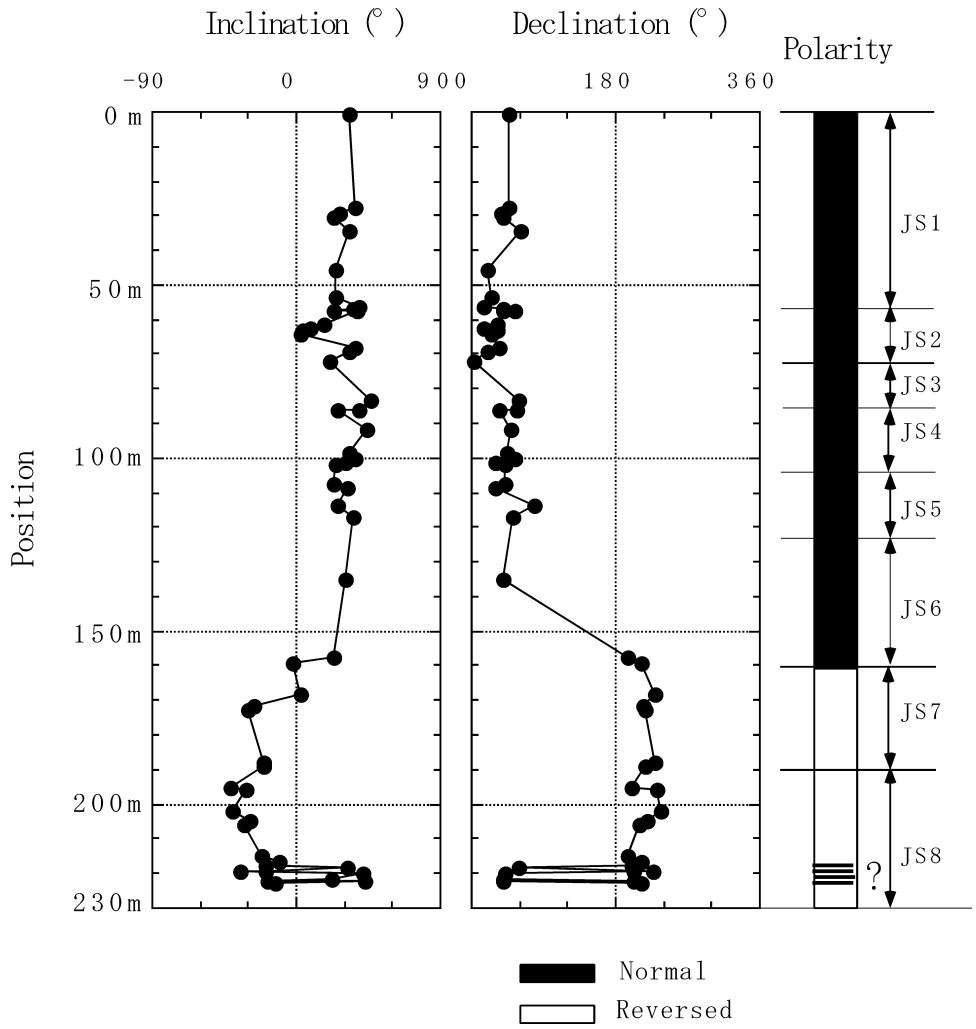



**Figure 6.** Magnetic polarity sequence of the Middle Triassic Huangmaqing Formation for the Nanjing area. Palaeomagnetic inclinations and declinations of the studied specimens are plotted versus their stratigraphic positions. Inclination and declination data of those 57 specimens, whose remanence vectors revealed a linear decay to the origin, are calculated by principal component analysis. Specimens are grouped into eight sites as JS1 to JS8.

components. A high-temperature component, which generally carries 50 per cent of the NRM intensity, was isolated between 670 and 690 °C. This component was identified in 15 out of 20 specimens, and its mean direction was calculated at the specimen level ($n = 15$ specimens): $D = 317.0°$, $I = 41.3°$, $k = 13.5$, $\alpha_{95} = 10.8°$ (*in situ*) and $D = 292.7°$, $I = 49.7°$, $k = 15.5$, $\alpha_{95} = 10.0°$ (tilt corrected).

## 6 DISCUSSION

Characteristic remanent directions (ChRM) have been successfully identified in the Middle Triassic red beds of all four areas studied in a temperature range of 610–690 °C. The pre-folding origin of palaeomagnetic directions for the red beds of the Badong Formation (Nanzhang area) is ascertained through positive fold and reversal tests. Although the fold tests are hindered by the monoclinal attitude of the red bed strata in the Tongling (Tongtoujian Formation) and Nanjing areas (Huangmaqing Formation), the tilt test (Enkin 2003) gives a maximum grouping of directions at 129 per cent and 93 per cent unfolding, respectively. The dual polarity directions obtained from these two formations passed a reversal test at the 95 per cent significance level. Based on these observations a pre-folding origin can be argued for the ChRM of the Tongling and Nanjing areas. In addition, the pattern of reversals obtained from the stratigraphic sequence of the Nanjing area (Fig. 6) is consistent with middle part of the Ladinian (as reported by Gallet *et al.* 1998), suggesting a primary origin for the ChRM component.

The palaeomagnetic results presented here can thus be incorporated into a reliable Middle Triassic data set available from the SCB (Table 2). Comparison of our results with the available Middle Triassic data will definitely contribute to a better understanding of intracontinental deformation in the SCB.

### 6.1 Stable belt along the northern margin of the SCB

From the northern margin of the SCB, a northeasterly declination of about 40° is observed in the Middle Triassic rocks of the Nanzhang, Tongling and Nanjing areas. These areas are located at a distance between 70 and 100 km from the Qinling-Dabie-Tan-Lu zone (Figs 1 and 8). The Nanzhang area, which gave a declination value of 38.2° ± 6.4°, is located at a distance of 90 km from the Xiangfan-Guangji fault zone of the Qinling-Dabie orogenic belt. Huang & Opdyke (1997) reported a declination of 38.5° ± 6.4° from the Badong Formation of the Jingmen-Dangyang Basin (31.4°N, 111.7°E), which is located about 100 km from the Xiangfan-Guangji

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**Table 2.** Triassic palaeomagnetic results from the South China Block. Key: No., numbers denoting data in Fig. 8; Fei, Feixianguen; T, Triassic; P, Permian; 1, Early; 2, Middle, *N*, *n*, number of localities (samples) used to determine palaeomagnetic direction or pole; Dec. and Inc., declination and inclination; $\alpha_{95}$, radius of the cone of 95 per cent confidence; Lat., Lon., latitude and longitude; $A_{95}$, the radius that the mean pole lies within 95 per cent confidence dp and dn are radii of the semi-axis of the ellipse of 95 per cent confidence along the site-to-pole great circle and perpendicular to the great circle, respectively. Area LP (Steiner *et al.* 1989) is located near the village of Liangfenya (106.4°E, 29.6°N). Since the palaeomagnetic direction from the Nanjing area is revised in this paper, the preliminary Middle Triassic data from the Nanjing area (Jiangsu; 12*) are not incorporated in the calculation of the mean. The distances in brackets for Nos 1–4 are distances from the Tan-Lu Fault.

| No | Area | Formation | Age | *N* | *n* | Locality | | Reference | Tilt corrected | | $\alpha_{95}$ | Pole position | | $A_{95}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | °E | °N | | Dec. (°) | Inc. (°) | | Lat. (°) | Lon. (°) | dp/dm |
| Areas close to SCB and NCB collision zone | | | | | | | | | | | | | | |
| 1 | Yueshan C (25 km) | Tongtoujian | Tr2 | 6 | | 116.47 | 30.23 | Gilder *et al.* (1999) | 272.0 | 32.9 | 13.0 | 10.8 | 41.3 | 13.2 |
| 2 | Yueshan (25 km) | Huangmaqing | Tr2 | | 15 | 116.81 | 30.61 | This study | 292.7 | 49.7 | 10.0 | 33.0 | 45.6 | 8.9/13.3 |
| 3 | Yueshan B (25 km) | Tongtoujian | Tr2 | | 17 | 116.83 | 30.64 | Gilder *et al.* (1999) | 280.8 | 34.7 | 5.9 | 19.1 | 39.1 | 3.9/6.4 |
| 4 | Yueshan A (33 km) | Tongtoujian | Tr2 | 9 | | 116.93 | 30.64 | Gilder *et al.* (1999) | 12.3 | 33.6 | 4.1 | 73.5 | 251.4 | 3.9 |
| Peripheral region | | | | | | | | | | | | | | |
| 5 | Emeishan (Lianghek) | Feixianguan | Tr1 | | 101 | 103.4 | 29.6 | Ma *et al.* (1993) | 27.7 | 15.6 | 3.3 | 56.2 | 227.6 | 1.7/3.4 |
| 6 | Shangsi | Feixianguan | P2/Tr1 | | 114 | 105.5 | 32.0 | Heller *et al.* (1988) | 35.9 | 12.8 | 5.3 | 47.9 | 225.1 | 2.8/5.0 |
| 7 | GPW (Shangsi) | Feixianguan | P/Tr | | 38 | 106.4 | 32.4 | Steiner *et al.* (1989) | 36.4 | 12.2 | 4.1 | 47.2 | 226.3 | 2.1/4.2 |
| 8 | Jingme-Dagyang | Badong | Tr2 | 18 | | 111.7 | 31.4 | Huang & Opdyke (1997) | 38.5 | 26.0 | 6.4 | 50.5 | 218.8 | 5.8 |
| 9 | Nanzhang | Badong | Tr2 | 21 | | 111.7 | 31.5 | This study | 37.6 | 25.3 | 6.2 | 51.0 | 220.9 | 3.6/6.7 |
| 10 | Tongling | Tongtoujian | Tr2 | 5 | | 117.9 | 30.9 | This study | 32.1 | 25.2 | 9.7 | 55.6 | 231.8 | 5.6/10.4 |
| 11 | Nanjing | Huangmaqing | Tr2 | 8 | | 118.9 | 32.0 | This study | 39.4 | 27.8 | 5.4 | 50.2 | 225.2 | 3.2/5.9 |
| 12 | Jiangsu (Nanjing) | Huangmaqing | Tr2 | | 23 | 119.0 | 32.0 | Opdyke *et al.* (1986) | 44.8 | 24.4 | 9.3 | 44.8 | 223.6 | 11.5 |
| 13 | Changxing | Yinkeng | Tr1 | | 16 | 119.7 | 31.0 | Dobson *et al.* (1993) | 35.7 | 22.6 | 11.2 | 51.8 | 232.1 | 11.2 |
| Mesozoic fold and thrust belt | | | | | | | | | | | | | | |
| 14 | Nanxi | Feixianguan | Tr1 | | 15 | 104.9 | 28.9 | Enkin *et al.* (1992a) | | | | 43.3 | 211.7 | 3.3/6.4 |
| 15 | Xuyong | Jianlingjiang/Fei | Tr1 | 15 | | 105.5 | 28.0 | Yang & Besse (2001) | 42.1 | 10.2 | 6.8 | 44.1 | 217.2 | 4.9 |
| 16 | LP (Liangtenya) | Feixianguan | P/Tr | | 17 | 106.4* | 29.6 | Steiner *et al.* (1989) | 37.4 | 6.2 | | 45.8 | 225.2 | 3.5/7.0 |
| 17 | HPT (Henchuan) | Feixianguan | Tr1 | | 62 | 106.3 | 29.9 | Steiner *et al.* (1989) | 50.4 | 17.1 | | 38.5 | 209.8 | 1.7/3.3 |
| 18 | HT (Chonggin) | Feixianguan | Tr1 | | 26 | 106.3 | 29.9 | Steiner *et al.* (1989) | 49.4 | 14.9 | | 38.7 | 211.8 | 3.2/6.3 |
| 19 | Tongzi (Guizhou) | Yelong | Tr1 | 9 | | 106.85 | 28.58 | Opdyke *et al.* (1986) | 40.0 | 13.1 | 11.0 | 46.3 | 219.2 | 10.9 |
| 20 | Changshou | Jianlingjiang/Fei | Tr1 | 5 | | 107.0 | 29.9 | Enkin *et al.* (1992b) | 46.1 | 11.9 | 8.7 | | | |
| 20 | Changshou | Jianlingjiang | | | 36 | 107.0 | 29.9 | Enkin *et al.* (1992a) | | | | 39.8 | 216.2 | 2.7/5.3 |
| 21 | Wulong+Shuijiang | Daye/Fei | Tr1 | | 88 | 107.3–107.7 | 29.25 | Heller *et al.* (1995) | 42.8 | 14.7 | 3.9 | 44.3 | 217.4 | 3.5 |
| 22 | Daye | Daye | Tr1 | | 15 | 110.0 | 29.5 | Dobson & Heller (1993) | 52.5 | 28.5 | 5.7 | 39.8 | 205.1 | 5.7 |
| 23 | Jianlingliang | Jianlingjiang | Tr1-2 | | 14 | 110.0 | 29.5 | Dobson & Heller (1993) | 62.8 | 40.6 | 8.2 | 34.0 | 191.1 | 8.2 |
| 24 | Badong | Badong | Tr2 | 16 | | 110.4 | 31.0 | Huang & Opdyke (1996) | 77.6 | 23.0 | 6.8 | 16.6 | 195.8 | 5.5 |
| 25 | Sangzhi | Badong | Tr2 | 20 | | 110.2 | 29.4 | Huang & Opdyke (1996) | 63.6 | 26.8 | 4.3 | 29.8 | 200.0 | 4.0 |
| 26 | Puxi | Puxi | Tr2 | 7 | | 113.9 | 29.7 | Huang & Opdyke (1996) | 55.1 | 14.2 | 7.5 | 33.8 | 215.8 | 5.8 |
| Other areas | | | | | | | | | | | | | | |
| | Zhangfeng | | Tr | 5 | | 105.4 | 25.3 | Wang *et al.* (1993) | 355.0 | 13.0 | 5.2 | 70.0 | 300.0 | 5.2 |
| | Guangxi-Jinji+Yufen | Baifeng | Tr2 | 7 | | 107.2 | 23.6 | Gilder *et al.* (1995) | 56.1 | 6.9 | 9.8 | 32.2 | 208.7 | 7.9 |
| | Fujiang | Xikou | Tr1 | | 18 | 117.2 | 26.0 | Gilder *et al.* (1995) | 302.7 | −25.2 | 6.0 | 19.4 | 2.1 | 3.8 |

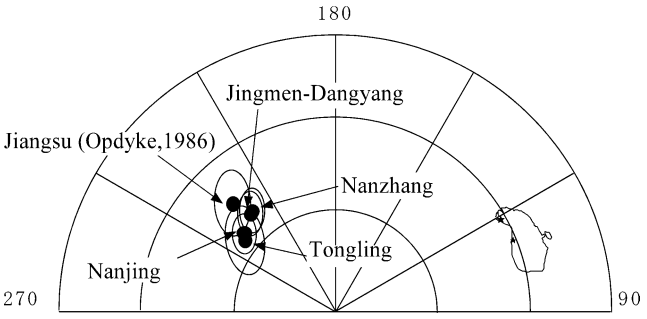



**Figure 7.** Equal area projection of the Middle Triassic palaeomagnetic poles (with associated circles of 95 per cent confidence level) from the northern margin of the SCB (Tongling, Nanjing and Nanzhang; the present study, Jiagmeng-Dangyang; Huang & Opdyke 1997). The numbers adjacent to poles represent the corresponding localities in Table 2.

Fault. The Tongling and Nanjing areas with declinations of 32.1° ± 9.7° and 34.9° ± 5.4° are at a distance of 75 and 70 km from the Tan-Lu Fault, respectively.

The corresponding Middle Triassic palaeomagnetic poles for the above mentioned four areas are compared in Fig. 7 and listed in Table 2 as nos 8–11. These four poles are clustered around $\lambda = 52.8°$N, $\phi = 224.7°$E. The coherence among these poles, which represent an area located between 70 and 100 km from the northern boundary, appears to be excellent (where individual poles are intersecting each other's confidence ellipses at the 95 per cent limit). This type of behaviour indicates that no significant relative tectonic rotations occurred among these four areas since the Middle Triassic. Although these four areas are sporadically located over a distance of 550 km along the northern margin of the SCB (extending from the Nanjing area at 32.0°N, 118.9°E to the Nanzhang area at 31.5°N, 111.7°E), the lack of relative tectonic rotation kept them as part of the stable northern belt.

The existence of a stable belt is also supported by four other palaeomagnetic directions reported from Permo-Triassic and Early Triassic rocks of the northwestern and eastern margins of the SCB (nos 5, 6, 7 and 13 in Table 2). These directions from Emeishan, Shangsi, GPW and Changxing areas are located between 75 km (29.6°N, 103.4°E ) and 225 km (31.0°N, 119.7°E) from the northern continental margin. The Late Permian–Early Triassic palaeomagnetic directions from these areas are characterized by 35° declination, while their palaeomagnetic poles are clustered around a mean pole position of $\lambda = 50.8°$N, $\phi = 227.7°$E, $A_{95} = 5.2°$ ($N = 4$). Tight clustering in the distribution of these poles also indicates that no relative rotation occurred between these areas. Coherent distribution of eight palaeomagnetic directions (including four previously reported Permo-Triassic to Early Triassic directions and the present four Middle Triassic directions) allows us to extend this stable belt up to 1800 km between 103.4°E and 119.7°E along the northern margin of the SCB (Fig. 8).

Discovery of the longitudinally wide stable belt does not support the idea of Gilder *et al.* (1999) that the northern margin of the SCB underwent oroclinal bending-type deformation as a result of continuous indentation of the NCB at the Dabieshan corner. As a first-order approximation, we strongly suggest that the SCB has maintained its present configuration at least from the time of the Middle Triassic.

### 6.2 Intracontinental deformation within the SCB

As shown in Fig. 8, clockwise rotation of the interior part of the SCB with respect to northern stable belt is evident from Permo-Triassic to Middle Triassic palaeomagnetic directions available from the Mesozoic fold-thrust belt (MFTB) and the northern studied areas. Larger easterly declinations (between 37° and 78°) from the MFTB are clearly in contrast with relatively smaller declinations (between 28° and 39°) from the northern stable belt. A palaeomagnetic pole for the whole MFTB is compared with that from the northern stable belt to evaluate the amount of relative rotation. As shown in Fig. 8b, the mean palaeomagnetic pole calculated from Permo-Triassic, Early Triassic and Middle Triassic results (nos 14–26 in Table 2) of the MFTB ($\lambda = 38.5°$N, $\phi = 209.8°$E, $A_{95} = 5.8°$) occupied different positions with respect to the contemporaneous pole of the northern stable belt ($\lambda = 51.8°$N, $\phi = 226.2°$E, $A_{95} = 3.1°$). Comparison of these two mean poles indicates that the MFTB was subjected to clockwise rotation by 17.7° ± 4.9° (around the reference position of 30°N, 110°E) with respect to the northern stable belt. Here, the 95 per cent confidence limit is obtained using the formulation of Debiche & Watson (1995).

Since a larger easterly declination is observed in the eastern part of the MFTB (nos 22–26 in Table 3 and Fig. 8) than in the western part, the Qirue Fault may have caused this divergence in palaeomagnetic directions. Mean palaeomagnetic poles calculated for western and eastern regions are $\lambda = 42.7°$N, $\phi = 215.9°$E, $A_{95} = 3.3°$ ($N = 8$) and $\lambda = 31.1°$N, $\phi = 201.3°$E, $A_{95} = 11.3°$ ($N = 5$), respectively. Clockwise rotation of 11.5° ± 5.0° is estimated for the western region of the MFBT (around 30°N, 103°E) with respect to northern stable belt even if it could be resulted from apparent polar wander between the Permo-Triassic and middle-Triassic. The most striking result, however, is indicated by the relative clockwise rotation of 30.3° ± 9.2° between the eastern part of the MFTB and the northern stable belt. This clockwise rotation is evaluated (Fig. 8c) by comparison of the mean Middle Triassic pole from the eastern part of the MFTB ($\lambda = 28.9°$N, $\phi = 200.5°$E, $A_{95} = 14.1°$) with that from northern stable belt ($\lambda = 52.8°$N, $\phi = 224.7°$E, $A_{95} = 5.2°$). We thus conclude that the MFBT as a whole experienced clockwise rotation of more than 11.5° ± 5.0° with respect to the northern stable belt later than the Middle Triassic, while its eastern part experienced an additional clockwise rotation by 19°.

The structural trend of the present MFBT is characterized by multilayer thrusts of Jurassic age (Hsu *et al.* 1990; Yan *et al.* 2003). Rotational motion within the MFBT is most probably associated with the formation of this structure. The Mesozoic sedimentary cover, which consists of the multilayer thrust structure, could have been rotated during detachment of the Palaeozoic–Mesozoic layers from the Proterozoic basement. Since a thick-skinned structure is predominant in the eastern part of the MFBT and a thin-skinned in its western counterpart (Yan *et al.* 2003), a divergent clockwise rotational motion in these regions may have been controlled for the reason explained.

### 6.3 Deformation near the Tan Lu Fault

Based on the Middle Triassic palaeomagnetic results, westerly declinations are observed within a zone of 33 km from the Tan-Lu Fault (Fig. 1). The results presented here from Yueshan area, which is located at a distance of 25 km from the fault zone, show a declination of 292.7°. Gilder *et al.* (1999) reported large westerly declinations of 280.8° and 272.0° at localities B and C of the Middle Triassic Tongtoujian Formation (25 km from the fault zone), respectively. However, a northerly declination of 12.3° has been reported at locality A (33 km from the fault) of the same formation. This type

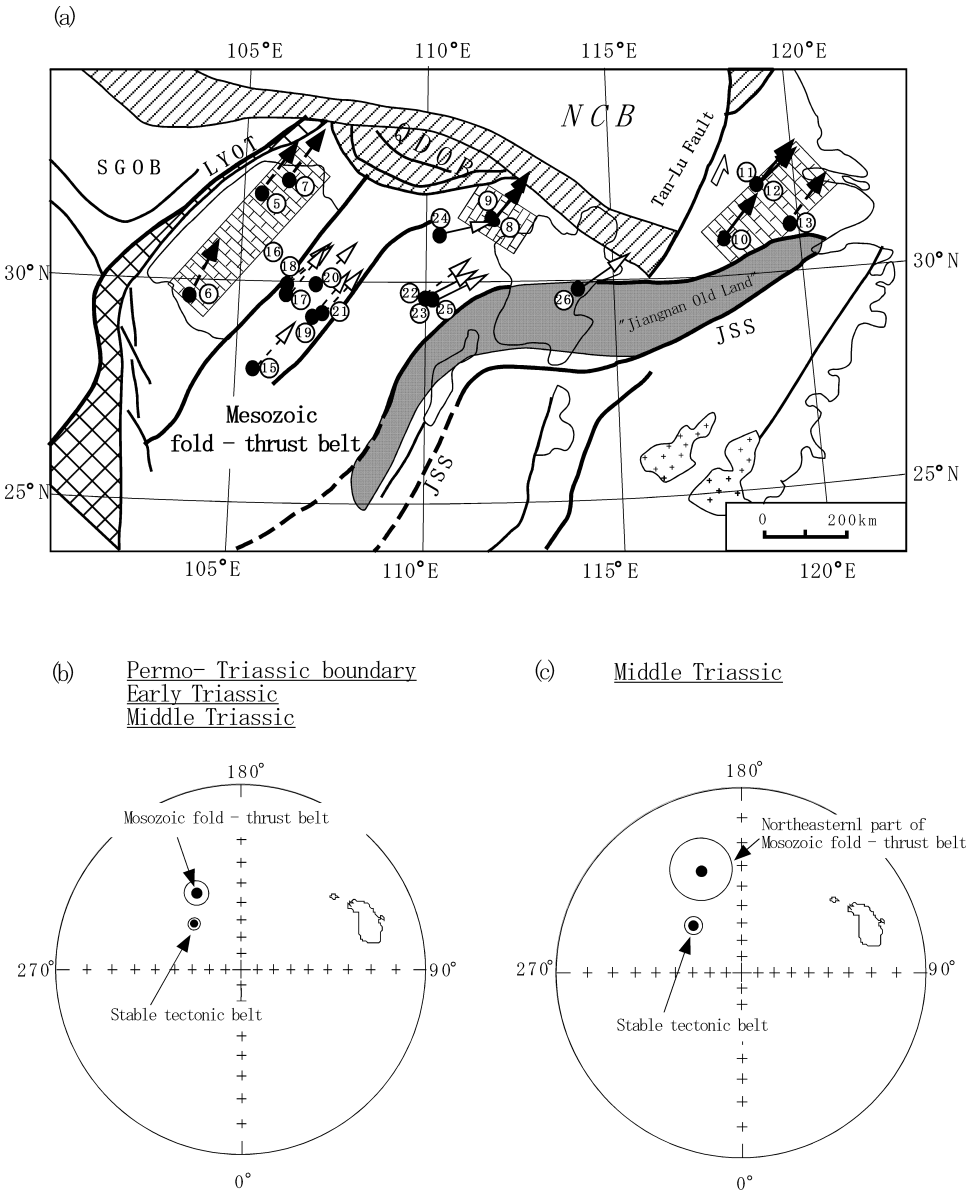



**Figure 8.** (a) Declinations of the Triassic palaeomagnetic directions from the northern stable belt and the Mesozoic fold-thrust belt in the SCB. Solid arrows are declinations from the northern stable tectonic belt. Open arrows are declinations from the Mesozoic fold-thrust belt. Solid axis of arrow shows Middle Triassic and dashed axis of arrow Permo-Triassic and Early Triassic. The numbers adjacent to the arrows represent the corresponding localities in Table 2. For explanation of the abbreviations see the caption to Fig 1. The Mesozoic fold-thrust belt is divided into the western (W) and eastern (E) regions by the Qirue Fault. (b) Comparison of the Permo-Triassic to Middle Triassic mean pole for the whole Mesozoic fold-thrust belt with that for northern stable belt of the SCB. (c) Comparison of the mean Middle Triassic pole for northeastern part of the Mesozoic fold-thrust belt with that of northern stable tectonic belt.

**Table 3.** Mean palaeomagnetic pole positions from the stable belt and the Mesozoic fold-thrust belt (MFTB). Key: $N$, number of localities used for the calculation of mean palaeomagnetic pole; Lat., Lon., latitude and longitude; $A_{95}$, the radius that the mean pole lies within 95 per cent confidence.

| Age | $N$ | Pole position | | $A_{95}$ | Reference nos (from Table 2) |
|---|---|---|---|---|---|
| | | Lat. (°) | Lon. (°) | | |
| Tectonic stable belt | | | | | |
| Tr2 | 4 | 52.8 | 224.7 | 5.1 | 8–11 |
| P2/ Tr1, Tr1 | 4 | 50.8 | 227.7 | 5.2 | 5–7, 13 |
| P2/Tr1, Tr1, Tr2 | 8 | 51.8 | 226.2 | 3.1 | 5–11, 13 |
| Mesozoic fold-thrust belt | | | | | |
| Whole belt: P2/Tr1, Tr1, Tr2 | 13 | 38.5 | 209.8 | 5.8 | 14–26 |
| Western region: P2/Tr1, Tr1 | 8 | 42.7 | 215.9 | 3.3 | 14–21 |
| Eastern region: Tr2 | 4 | 28.9 | 200.5 | 14.1 | 23–26 |
| Eastern region: Tr1, Tr2 | 5 | 31.1 | 201.3 | 11.3 | 22–26 |

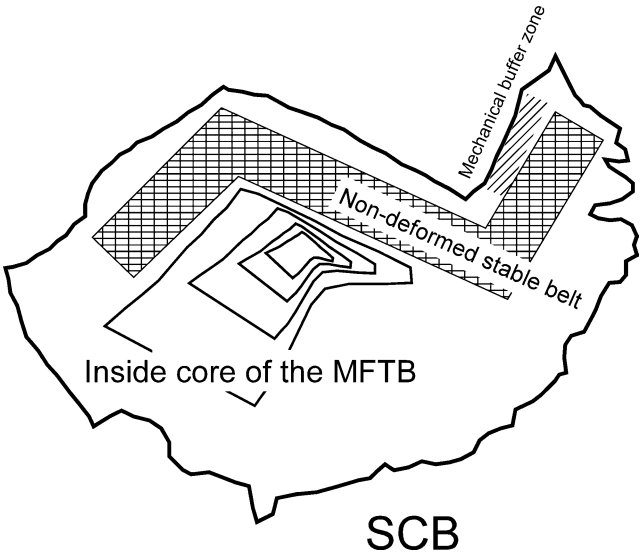



**Figure 9.** Tectono-structural model of the SCB. The SCB consists of three tectonic zones from continental margin to interior, i.e. a mechanical buffer zone, the non-deformed stable belt and the inside core of the MFTB (from a tectonic rotation point of view). A mechanical buffer zone is characterized by a narrow belt located within 33 km of the northern margin of the SCB, in which a crustal portion is subjected to up to 90° clockwise rotation. The stable tectonic belt is located between 70 and 225 km from the northern SCB boundary, in which no relative rotational motion has occurred since the Middle Triassic. The inside core of the Mesozoic fold-thrust belt (MFTB) experienced a clockwise rotation of more than 11.5° ± 5.0° with respect to northern stable belt.

of behaviour thus indicates that areas close to Tan-Lu Fault reveal larger westerly declinations than the interior areas.

Westerly declinations of the peripheral region are in contrast to the northeasterly declinations of the northern stable belt. Since the Tan-Lu Fault forms a boundary between the SCB and NCB, tectonic interaction between these two blocks during and after their collision brought about a counter-clockwise rotation in the area close to the fault zone. Left lateral shear along the Tan-Lu Fault (e.g. Xu *et al.* 1987) is probably responsible for a large counter-clockwise rotation. This phenomenon can be compared with a 50 km zone across the Indus-Zangbo Suture, where a collision between India and Asia has resulted in more than 80° counter-clockwise rotation (Pozzi *et al.* 1984; Otofuji *et al.* 1989, 1991). Adjacent regions on both side of the collisional boundary (intercontinental scale) provide a mechanical buffer zone, where tectonic related stresses can possibly be developed.

### 6.4 Some remarks about the suturing process between the SCB and the NCB

Almost similar Middle Triassic pole positions from Nanzhang, Tongling, Nanjing and the Jingmen-Dangyang Basin (northern margin of the SCB) provide a solid reason to count these areas as part of a single autochthonous block. Comparison of these palaeopoles with the Middle Triassic pole of the NCB (Yang & Besse 2001) indicates an angular difference of 65.8° ± 6.8°, suggesting that final suturing between these two blocks occurred after the Middle Triassic. The post-Middle Triassic commencement of suturing between these two blocks is in good agreement with the 240–210 Ma exhumation event of ultra-high-pressure metamorphism in the Dabie-Sulu area of eastern China (Ames *et al.* 1993; Webb *et al.* 1999; Hacker *et al.* 2000). In addition, the post-Middle Triassic suturing between the SCB and the NCB was associated with relative rotation of more than 60°.

Several models of suturing between the NCB and SCB have been reported (e.g. Ratschbacher *et al.* 2003), but no significant deformation of the northern margin of the SCB has been mentioned. Although our palaeomagnetic results from the Nanjing and Tongling areas (on the eastern side of the Tan-Lu Fault) support the idea of irregular geometry along the northeastern margin of the SCB, which may be extended up to 550 km farther north than its western counterpart (Yin & Nie 1993). The original configuration of the northern SCB is, however, still an enigma, and could be similar to that of Greater India (e.g. Klootwijk *et al.* 1979, 1985). Yin & Nie (1993) have suggested that an indentation of irregular northern margin of the SCB into the NCB created the Tan-Lu fault zone in eastern China and the Honam shear zone in South Korea, but this idea seems handicapped by the fact that a relative counter-clockwise rotation of 60° has been reported between these two blocks since the Middle Triassic. This relative rotational movement might have resulted in an extensional tectonic activities in the Tan-Lu fault zone. It thus seems impossible that commencement of tectonic activities along the Tan-Lu fault zone took place prior to complete suturing of the SCB and NCB, e.g. the early-Middle Jurassic (Gilder *et al.* 1999; Yang & Besse 2001).

We therefore conclude that any future modelling regarding the initial collision and subsequent suturing of the NCB with the SCB must deal with large relative rotational movement and coeval exhumation of the ultra-high-pressure metamorphic rocks. Post-Early Jurassic tectonic activities along the Tan-Lu Fault have to be taken into consideration, and could be linked to post-collision shortening between these two megablocks.

## 7 CONCLUSION

The available Permo-Triassic to Middle Triassic palaeomagnetic directions divides the SCB into three tectonic zones. From the continental margin to the interior these zones are (1) a mechanical buffer zone of shear stresses close to the boundary, (2) a relatively stable belt characterized by non-oroclinal bending and (3) the inside core of the MFTB (Fig. 9).

Large post-Triassic clockwise rotation (up to 90°) has been detected within a zone of 33 km from the Tan-Lu Fault. This peripheral region along the northern margin of the SCB behaves as a mechanical buffer zone where shear stresses were brought about during the suturing process between the SCB and the NCB. Because of their absorption in a peripheral deformed region, very little shear stress invaded the adjacent stable belt.

The relatively stable belt along the northern margin of the SCB extends from Changxing, through Nanjing, Tongling and Nanzhang, to Emeishan. This belt is located between 70 and 225 km from the northern margin of the SCB, in which no relative rotational motion occurred after the Middle Triassic. The inside core of the MFTB which occupies the interior part of the SCB experienced a clockwise rotational motion of 11.5° ± 5.0° to 30.3° ± 9.2° with respect to the northern stable belt later than the Middle Triassic.

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